www.Padasalai.Net www.TrbTnpsc.com

COMMON QUARTERLY EXAMINATION – 2026

P

Standard XII

Reg.No. :

MATHEMATICS

Part – I

Time: 3.00 hrs. Marks: 90

20 x 1 = 20

I. Choose the correct answer:

1. If $A\begin{bmatrix}1 & -2\\ 1 & 4\end{bmatrix}=\begin{bmatrix}6 & 0\\ 0 & 6\end{bmatrix}$, then A =

a) $\begin{bmatrix}1 & -2\\ 1 & 4\end{bmatrix}$ b) $\begin{bmatrix}1 & 2\\ -1 & 4\end{bmatrix}$ c) $\begin{bmatrix}4 & 2\\ -1 & 1\end{bmatrix}$ d) $\begin{bmatrix}4 & -1\\ 2 & 1\end{bmatrix}$

2. A is of order n, $\lambda \neq 0$, then adj(λA) =

a) λ^{n-1} adj(A) b) λ^{n-2} adj(A) c) λ^{n} adj(A) d) $\frac{1}{\lambda}$ adj(A)

3. If $0 \leq \theta \leq \pi$ and the system of equations
$x + (\sin\theta)\, y - (\cos\theta)\, z = 0$, $(\cos\theta)\, x - y + z = 0$, $(\sin\theta)\, x + y - z = 0$
has a non–trivial solution then θ is

a) $\frac{2\pi}{3}$ b) $\frac{3\pi}{4}$ c) $\frac{5\pi}{6}$ d) $\frac{\pi}{4}$

4. If $z=\frac{(\sqrt{3}+i)^3\,(3i+4)^2}{(8-6i)^2}$ then |z| is equal to

a) 0 b) 1 c) 2 d) 3

5. If $|z_1| = 1$, $|z_2| = 2$, $|z_3| = 3$ and $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$, then the value of $|z_1 + z_2 + z_3|$ is

a) 1 b) 2 c) 3 d) 4

6. If α and β are the roots of $x^2 + x + 1 = 0$, then $\alpha^{2020} + \beta^{2020}$ is

a) –2 b) –1 c) 1 d) 2

7. The number of real numbers in $[0,2\pi]$ satisfying $\sin^4 x - 2\sin^2 x + 1$ is

a) 2 b) 4 c) 1 d) ∞

8. If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \frac{3\pi}{2}$, the value of $x^{2017} + y^{2018} + z^{2019} - \frac{9}{x^{101}+y^{101}+z^{101}}$ is

a) 0 b) 1 c) 2 d) 3

9. e^{ix} is a periodic function with period

a) 0 b) π c) 2π d) 4π

10. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is

a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$

11. If x + y = k is a normal to the parabola $y^2 = 12x$, then the value of k is

a) 3 b) –1 c) 1 d) 9

12. Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

a) 2ab b) ab c) $\sqrt{ab}$ d) $\frac{a}{b}$

Kindly Send Your Study Materials To Us Email ID: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

13. If the length of the perpendicular from the origin to the plane $2x + 3y + \lambda z = 1$, $\lambda > 0$ is $\frac{1}{5}$, then the value of λ is

a) $2\sqrt{3}$ b) $3\sqrt{2}$ c) 0 d) 1

14. If the line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ lies in the plane $x + 3y - \alpha z + \beta = 0$, then (α, β) is

a) (–5,5) b) (–6,7) c) (5,–5) d) (6,–7)

15. Distance from the origin to the plane $3x - 6y + 2z + 7 = 0$ is

a) 0 b) 1 c) 2 d) 3

16. If α, β and γ are the roots of $x^3 + px^3 + qx + r$, then $\sum \frac{1}{\alpha}$ is

a) $-\frac{q}{r}$ b) $-\frac{p}{r}$ c) $\frac{q}{r}$ d) $-\frac{q}{p}$

17. If A,B and C are invertible matrices of some order, then which one of the following is not true?

a) adj A = |A| A^{-1} b) adj(AB) = (adj A) (adj B)

c) det A^{-1} = (det A)$^{-1}$ d) $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$

18. If $\cot^{-1} 2$ and $\cot^{-1} 3$ are two angles of a triangle, then the third angle is

a) $\frac{p}{4}$ b) $\frac{3p}{4}$ c) $\frac{p}{6}$ d) $\frac{p}{3}$

19. For the parabola $(x - h)^2 = -4a(y - k)$ the equation of directrix is

a) y = k b) y = a c) y = k + a d) x = k + a

20. The non–parametric form of a vector equation passing through a points whose position vectors are $\vec{a}$ and parallel to two vectors $\vec{u}$ and $\vec{v}$ is

a) $[\vec{r} - \vec{u}, \vec{u}, \vec{v}] = 0$ b) $[\vec{r} - \vec{a}, \vec{u}, \vec{v}] = 0$

c) $[\vec{r} - \vec{v}, \vec{u}, \vec{v}] = 0$ d) $[\vec{r} - \vec{u}, \vec{a}, \vec{v}] = 0$

Part - II

II. Answer any 7 questions. (Q.No.30 is compulsory) **7 x 2 = 14**

21. If A is symmetric, prove that then adj A is also symmetric.

22. Form a polynomial equation with integer coefficients with $\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}$ as a root

23. Evaluate: $i^{59} + \frac{1}{i^{59}}$

24. Find the angle between the planes $\vec{r} \cdot (\hat{i} + \hat{j} - 2\hat{k}) = 3$ and $2x - 2y + z = 2$.

25. Simplify $\left(\sin\frac{\pi}{6} + i\cos\frac{\pi}{6}\right)^{18}$

26. Find the equation of the circle with centre (2,–1) and passing through the point (3,6) in standard form.

Kindly Send Your Study Materials To Us Email ID: padasalai.net@gmail.com

27. If $\cot^{-1}\left(\frac{1}{7}\right)=\theta$, find the value of $\cos\theta$

www.Padasalai.Net www.TrbTnpsc.com

28. Determine the number of positive and negative roots of the equation $x^9-5x^8-14x^7=0$.

29. Foci $(\pm2, 0)$, $e=\frac{3}{2}$, Find the equation of the hyperbola

30. If $2\vec{i}-\vec{j}+3\vec{k}$, $3\vec{i}+2\vec{j}+\vec{k}$, $\vec{i}+m\vec{j}+4\vec{k}$ are coplanar, find the value of m.

Part - III

III. Answer any 7 questions. (Q.No.40 is compulsory) **7 x 3 = 21**

31. An equation of the elliptical part of an optical lens system is $\frac{x^2}{16}+\frac{y^2}{9}=1$. The parabolic part of the system has a focus in common with the right focus of the ellipse. The vertex of the parabola is at the origin and the parabola opens to the right. Determine the equation of the parabola.

32. Solve : $12x^3+8x=29x^3-4$

33. Solve the systems of linear equations by Cramer's rule: $5x-2y+16=0$, $x+3y-7=0$

34. Prove that $\tan^{-1}x+\tan^{-1}\frac{2x}{1-x^2}=\tan\left(\frac{3x-x^2}{1-3x^2}\right)$, $|x|<\frac{1}{\sqrt{3}}$

35. Prove by vector method that an angle in a semi–circle is a right angle.

36. Find the inverse the non-singular matrix $A=\begin{bmatrix}0 & 5\\ -1 & 6\end{bmatrix}$, by Gauss-Jordan method.

37 Find the domain of $\sin^{-1}(2-3x^2)$.

38. A straight line passes through the point (1,2,–3) and parallel to $4\vec{i}+5\vec{j}-7\vec{k}$. Find vector equation in non–parametric form

39. If the normal at the point 't_1' on the parabola $y^2=4ax$ meets the parabola again at the point 't_2' then prove that $t_2=-\left(t_1+\frac{2}{t_1}\right)$.

40. If $|z|=1$, show that $2\le|z^2-3|\le4$.

Part - IV

IV. Answer all the questions. **7 x 5 = 35**

41. a) Solve the following system of linear equations, by Gaussian elimination method: $4x+3y+6z=25$, $x+5y+7z=13$, $2x+9y+z=1$.

(OR)

b) Show that the line $x-y+4=0$ is a tangent to the ellipse $x^2+3y^2=12$. Also find the coordinates of the point of contact.

42. a) Prove by vector method that $\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$.

(OR)

b) Prove that $\tan\left(\sin^{-1}x\right)=\frac{x}{\sqrt{1-x^2}}$, $-1<x<1$

Kindly Send Your Study Materials To Us Email ID: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

43. a) Find the vertex, focus, directrix, and length of the latus rectum of the parabola $y^2 + 10y - 4x + 29 = 0$.

(OR)

b) Find parametric form of vector equation and Cartesian equations of the plane passing through the points (2,2,1), (1,–2,3) and parallel to the straight line passing through the points (2,1,–3) and (–1,5,–8) .

44. a) Show that

i) $\left(2+i\sqrt{3}\right)^{10} - \left(2-i\sqrt{3}\right)^{10}$ is purely imaginary

ii) $\left(\frac{19-7i}{9+i}\right)^{12} + \left(\frac{20-5i}{7-6i}\right)^{12}$ is real

(OR)

b) Solve : $\cos\left(\sin^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)\right) = \sin\left\{\cot^{-1}\left(\frac{3}{4}\right)\right\}$

45. a) Investigate the values of λ and μ the system of linear equations $2x + 3y + 5z = 9$, $7x + 3y - 5z = 8$, $2x + 3y + \lambda z = \mu$, have

i) no solution

ii) a unique solution

iii) an infinite number of solutions.

(OR)

b) A tunnel through a mountain for a four lane highway is to have a elliptical opening. The total width of the highway (not the opening) is to be 16m, and the height at the edge of the road must be sufficient for a truck 4m high to clear if the highest point of the opening is to be 5m approximately. How wide must the opening be?

46. a) Solve the equation $z^3 + 8i = 0$

(OR)

b) If $2 + i$ and $3 - \sqrt{2}$ are roots of the equation

$x^6 - 13x^5 + 62x^4 - 126x^3 + 65x^2 + 127x - 140 = 0$ find all roots.

47. a) Solve the following equation: $x^2 - 10x^3 + 26x^2 - 10x + 1 = 0$.

(OR)

b) If $\vec{a} = 2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{b} = 3\hat{i} + 5\hat{j} + 2\hat{k}$, $\vec{c} = -\hat{i} - 2\hat{j} + 3\hat{k}$ verify that

$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a}.\vec{c})\vec{b} - (\vec{a}.\vec{b})\vec{c}$

****

Kindly Send Your Study Materials To Us Email ID: padasalai.net@gmail.com